

IFS 244 EXERCISES WEEK 3 SEPTEMBER 2026

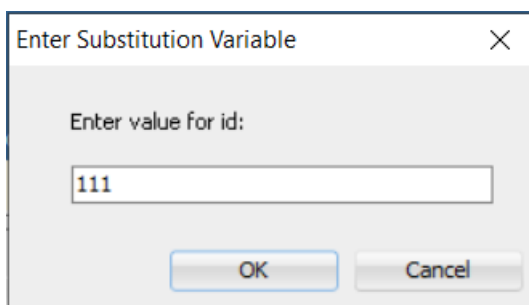
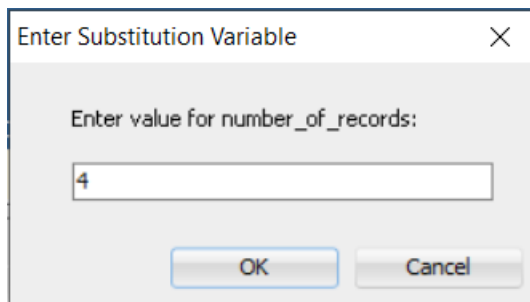
These exercises must be completed by 12h00 Monday 21 September 2026 and submitted to the tutors for marking on or before that time.

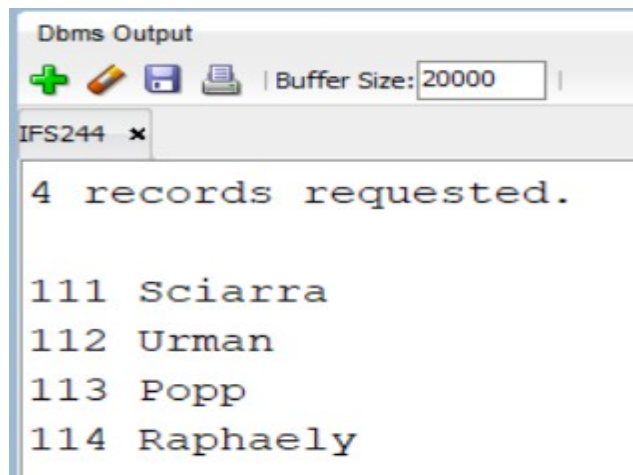
Ex. 1

Write a PL/SQL routine in SQL Developer that will solicit the number of records required from the user, by using an input dialogue, and then produce output in the DBMS Output window. The output must begin by saying the number of records requested, and follow with the employee ID and last name of each employee; the starting employee ID must also be solicited from the user in an input dialogue. To produce the records, the routine must find records sequentially from the ID entered by the user, until the required number of records has been produced. To produce an input dialogue in PL/SQL you must declare a variable to hold the input data and then use the '&' syntax to assign the input value to the variable:

```
v_emp_name := '&Employee Name:';
```

The text 'Employee name' above will be shown as the prompt in the dialogue. Here are images of the required inputs for this problem, in action:





```
Dbms Output
+ | Buffer Size: 20000 |
IFS244 x
4 records requested.

111 Sciarra
112 Urman
113 Popp
114 Raphaely
```

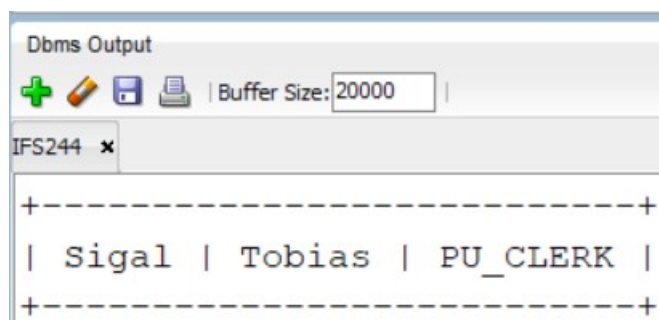
Your code must additionally handle a NO_DATA_FOUND exception by printing a suitable message giving the ID at which fetching a record failed, to the DBMS Output window – see the output at the top of the next page.

```
5 records requested.

205 Higgins
206 Gietz
No data was found from ID 207.
```

Ex. 2

Write a PL/SQL routine in SQL Developer that will solicit an employee ID, and declare variables to hold an employee first name, last name, and job ID. Run a query that selects from the EMPLOYEES table using the user-input ID to get data for the variables. The routine must then print the extracted data in the following format:



```
Dbms Output
+ | Buffer Size: 20000 |
IFS244 x
+-----+
| Sigal | Tobias | PU_CLERK |
+-----+
```

The purpose of this exercise is for you to create a program that will adjust itself to the different names and jobs produced by different employee IDs, and print them correctly inside a row as above, with the correct number of n-dashes for each set of values. Note that there is a space char before and after each string in the row.

In PL/SQL, to print just a single character to the output buffer, you must call DBMS_OUTPUT.PUT followed by DBMS_OUTPUT.NEW_LINE to flush the buffer and print the characters – see an example below:

```
BEGIN
  FOR i IN 1..5 LOOP
    DBMS_OUTPUT.PUT('X');
  END LOOP;
  DBMS_OUTPUT.NEW_LINE;
END;
```